INCH - POUND
MIL-PRF-32055
16 December 1999

PERFORMANCE SPECIFICATION

GUARD, SAFETY, MOBILE TIRE INFLATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification is for a mobile tire inflation safety guard (MTISG) used for inflating aircraft tires. The MTISG will accommodate tire sizes up to and including 37X11.5R18 with a regulated tire shop pressure of 360 psi. The MTISG will be capable of preventing the wheel components from being thrown outside or beyond the cage.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Standards. The following standard form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-595	Colors Used in Government Procurement

(Unless otherwise indicated, copy of the above standard is available from the Defense Automated Printing Service, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094, phone 215/697-2569)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be sent to SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Kelly AFB, TX 78241-6425 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910

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DEPARTMENT OF DEFENSE

AIR FORCE

AFOSH-STD-91-100 Air Force Occupational Safety and Health

(Unless otherwise indicated, copy of the above standard is available from Headquarters Air Force Safety Center, Ground Safety Division, Safety Engineering and Standards Branch (HQ AFSC/SEGS), 9700 Avenue G, SE, Suite 222, Kirkland AFB, NM 87117-5670, or http://afpubs.hq.af.mil .)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 General. Construction of the MTISG shall be durable enough to sustain the blast from the wheel tire assembly failure and shall prevent parts from single and multi-piece rims from flying out during a rim separation. The unit shall have forklift pockets as a minimum. During operation, the unit must demonstrate solid and stable contact with the shop floor. A rolling capability for ease of movement and equipment positioning is also required. The size and weight of the MTISG shall be minimized to enhance mobility. Maximum outside dimensions of the inflation cage shall not exceed 48 inch length X 48 inch height X 24 inch width. Minimum inside dimensions shall accommodate tire/wheel assembly roll in/roll out capability and required pressure dispersion during wheel tire assembly failure at proof test pressure.

3.3 Performance. The MTISG shall be capable of inflating tires up to and including the sizes of 37X11.5R18 with a regulated tire shop pressure of 360 psi. The MTISG shall have the capacity to withstand the maximum force that would be transferred to it during a wheel separation occurring at 150% of the maximum tire specification pressure for the type of wheel being serviced (AFOSH-STD-91-100). The unit shall be constructed so as to contain the debris emanated from a wheel tire assembly failure with a maximum rating of 540 psi. The unit shall be mobile.

3.4 Life. The overall operational life of the unit shall be designed for a minimum of twenty years.

3.5 Corrosion resistance. Measures will be taken such that the selection of materials, coatings and design features shall provide corrosion resistance using industry's best commercial practices. Dark green (FED-STD-595, color no. 24052) has been designated as the standard paint color for support equipment.

3.6 Maintainability. A minimum number of parts and tools shall be required for maintenance by design practices, such as reducing the variety of bolt head sizes to the practical minimum.

3.7 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle cost.

4. VERIFICATION

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4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (4.3)

b. Conformance inspection (4.4)

4.2 First article. The supplier shall furnish one MTISG for first article inspection. The first article shall conform to the requirements of this specification. The approved first article and the production unit shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

4.3 First article inspection. Unless otherwise specified, first article inspection shall consist of the examination and testing under section 4.0. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.3.1 Test sample. The test unit shall be representative of the production units the contractor intends to furnish.

4.3.2 Examination. Examination of the MTISG shall be conducted to conform adherence to the requirements herein. This examination shall consist of visual inspection of the MTISG and review of the design and manufacturing drawings. A functional check shall be made of the unit to insure that it is operational. The examinations shall be accomplished using a contractor-developed checklist that shall identify all specific attributes and characteristics for which inspection is made, how conformance is determined and results obtained.

4.3.3 Testing. The MTISG shall be tested to confirm adherence to the requirements herein. The first article will be tested to ensure that the unit will contain the debris emanated from a wheel tire assembly failure with a maximum psi rating of 540 psi. These tests shall be accomplished using contractor-developed procedures.

4.4 Conformance inspection. Unless otherwise specified, conformance inspection shall be applied to production items offered for acceptance under the contract and shall consist of examination of the units to ensure compliance with the requirements as stated in 3. Failure of any unit to pass any examination shall be cause for rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, those personnel need to contract the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

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(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. This item is used for inflating aircraft tires in a deployment scenario. There is no commercially available equipment that is mobile and capable of containing material that could be created by a 540-psi force generated from a tire explosion.

6.2 Acquisition requirements. Acquisition documents will specify the following:

a. Title, number and date of this specification.

b. When first article is required and quantity for testing (see 3.1 and 4.2).

c. Applicable levels of preservation, packaging, and packing required (see 5.1).

6.3 Part or Identifying Number (PIN). The PIN to be used for MTISG equipment acquired to this specification is created as follows:

M	32055
Prefix for military specification	Specification number

All items supplied per this specification will have the PIN “M32055” identity.

6.4 Subject term (key word) listing.

Aircraft tires
Blast
Debris
Explosion
Regulated
Separation
Wheel

Custodian:
Air Force - 99
Army - AL
DLA - CC

Preparing activity:
Air Force – 82

Project 4910-0823

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-32055	**2. DOCUMENT DATE (YYYYMMDD)** 19991216

3. DOCUMENT TITLE **GUARD, SAFETY, MOBILE TIRE INFLATION**

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME SA-ALC/TILDD	b. TELEPHONE *Include Area Code)* (1) Commercial (210) 925-6314 (2) AUTOVON 945-6314
c. ADDRESS *(Include Zip Code)* 485 QUINTEN ROOSEVELT ROAD KELLY AFB, TX 78241-6425	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533, Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

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